

Ответы на задание

```
[1/2] 50-cloud-init.yaml
```

```
network:
  renderer: networkd
  version: 2
  ethernets:
    enp0s3:
      dhcp4: true
      addresses: [192.168.1.18/24]
      gateway4: 192.168.1.1
      nameservers:
        addresses: [1.1.1.1, 8.8.8.8]
```

```
ubuntu@ubuntu:/etc/netplan$ sudo netplan --debug apply
** (generate:2433): DEBUG: 21:31:03.019: starting new processing pass

** (generate:2433): WARNING **: 21:31:03.021: `gateway4` has been deprecated, use default routes instead.
See the 'Default routes' section of the documentation for more details.
** (generate:2433): DEBUG: 21:31:03.022: We have some netdefs, pass them through a final round of validation
** (generate:2433): DEBUG: 21:31:03.024: enp0s3: setting default backend to 1
** (generate:2433): DEBUG: 21:31:03.024: Configuration is valid
** (generate:2433): DEBUG: 21:31:03.025: Generating output files..
** (generate:2433): DEBUG: 21:31:03.025: Open vSwitch: definition enp0s3 is not for us (backend 1)
** (generate:2433): DEBUG: 21:31:03.025: NetworkManager: definition enp0s3 is not for us (backend 1)
DEBUG:netplan generated networkd configuration changed, reloading networkd
DEBUG:Cannot call Open vSwitch: Cannot apply OVS cleanup: ovsdb-server.service is 'not-found'.
DEBUG:no netplan generated NM configuration exists
** (process:2431): DEBUG: 21:31:03.552: starting new processing pass

** (process:2431): WARNING **: 21:31:03.553: `gateway4` has been deprecated, use default routes instead.
See the 'Default routes' section of the documentation for more details.
** (process:2431): DEBUG: 21:31:03.553: We have some netdefs, pass them through a final round of validation
** (process:2431): DEBUG: 21:31:03.553: enp0s3: setting default backend to 1
** (process:2431): DEBUG: 21:31:03.558: Configuration is valid
DEBUG:Merged config:
b''
DEBUG:Link changes: {}
DEBUG:netplan triggering .link rules for lo
DEBUG:netplan triggering .link rules for enp0s3
** (process:2431): DEBUG: 21:31:03.691: starting new processing pass

** (process:2431): WARNING **: 21:31:03.695: `gateway4` has been deprecated, use default routes instead.
See the 'Default routes' section of the documentation for more details.
** (process:2431): DEBUG: 21:31:03.695: We have some netdefs, pass them through a final round of validation
** (process:2431): DEBUG: 21:31:03.695: enp0s3: setting default backend to 1
** (process:2431): DEBUG: 21:31:03.695: Configuration is valid
DEBUG:Merged config:
b''
ubuntu@ubuntu:/etc/netplan$ _
```

```
ubuntu@ubuntu:~$ ping -c 6 yandex.ru
PING yandex.ru (5.255.255.77) 56(84) bytes of data:
64 bytes from yandex.ru (5.255.255.77): icmp_seq=1 ttl=255 time=8.56 ms
64 bytes from yandex.ru (5.255.255.77): icmp_seq=2 ttl=255 time=8.41 ms
64 bytes from yandex.ru (5.255.255.77): icmp_seq=3 ttl=255 time=8.07 ms
64 bytes from yandex.ru (5.255.255.77): icmp_seq=4 ttl=255 time=8.01 ms
64 bytes from yandex.ru (5.255.255.77): icmp_seq=5 ttl=255 time=8.58 ms
64 bytes from yandex.ru (5.255.255.77): icmp_seq=6 ttl=255 time=6.64 ms

--- yandex.ru ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5005ms
rtt min/avg/max/mdev = 6.636/8.043/8.576/0.666 ms
ubuntu@ubuntu:~$ ping -c 6 skillbox.ru
PING skillbox.ru (185.65.148.253) 56(84) bytes of data:
64 bytes from 185.65.148.253: icmp_seq=1 ttl=255 time=4.33 ms
64 bytes from 185.65.148.253: icmp_seq=2 ttl=255 time=5.36 ms
64 bytes from 185.65.148.253: icmp_seq=3 ttl=255 time=5.24 ms
64 bytes from 185.65.148.253: icmp_seq=4 ttl=255 time=5.05 ms
64 bytes from 185.65.148.253: icmp_seq=5 ttl=255 time=6.68 ms
64 bytes from 185.65.148.253: icmp_seq=6 ttl=255 time=5.57 ms

--- skillbox.ru ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5005ms
rtt min/avg/max/mdev = 4.334/5.372/6.676/0.699 ms
ubuntu@ubuntu:~$
```

ping — команда для проверки доступности удалённого хоста в сети.

```
ubuntu@ubuntu:~$ traceroute -m 5 yandex.ru
traceroute to yandex.ru (5.255.255.77), 5 hops max
 1  10.0.2.2  0.003ms  0.019ms  0.004ms
 2  * * *
 3  * * *
 4  * * *
 5  * * *
ubuntu@ubuntu:~$ traceroute -m 5 vk.com
traceroute to vk.com (87.240.132.67), 5 hops max
 1  10.0.2.2  0.003ms  0.003ms  0.004ms
 2  * * *
 3  * * *
 4  * * *
 5  * * *
ubuntu@ubuntu:~$ _
```

traceroute — утилита для отслеживания маршрута, по которому пакеты данных проходят от компьютера до указанного удалённого хоста.

```
ubuntu@ubuntu:~$ dig yandex.ru

<> DiG 9.20.0-2ubuntu3.1-Ubuntu <> yandex.ru
;; global options: +cmd
;; Got answer:
;;->HEADER<- opcode: QUERY, status: NOERROR, id: 14249
;; flags: qr rd ra; QUERY: 1, ANSWER: 3, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;yandex.ru.                IN      A

;; ANSWER SECTION:
yandex.ru.                463     IN      A      77.88.44.55
yandex.ru.                463     IN      A      5.255.255.77
yandex.ru.                463     IN      A      77.88.55.88

;; Query time: 20 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Wed Apr 16 21:56:41 UTC 2025
;; MSG SIZE rcvd: 86

ubuntu@ubuntu:~$ dig @1.1.1.1 yandex.ru

<> DiG 9.20.0-2ubuntu3.1-Ubuntu <> @1.1.1.1 yandex.ru
; (1 server found)
;; global options: +cmd
;; Got answer:
;;->HEADER<- opcode: QUERY, status: NOERROR, id: 26594
;; flags: qr rd ra; QUERY: 1, ANSWER: 3, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
;; QUESTION SECTION:
;yandex.ru.                IN      A

;; ANSWER SECTION:
yandex.ru.                232     IN      A      77.88.44.55
yandex.ru.                232     IN      A      77.88.55.88
yandex.ru.                232     IN      A      5.255.255.77

;; Query time: 13 msec
;; SERVER: 1.1.1.1#53(1.1.1.1) (UDP)
;; WHEN: Wed Apr 16 21:56:42 UTC 2025
;; MSG SIZE rcvd: 86

ubuntu@ubuntu:~$ _
```

```
ubuntu@ubuntu:~$ dig google.ru

<> DiG 9.20.0-2ubuntu3.1-Ubuntu <> google.ru
;; global options: +cmd
;; Got answer:
;;->HEADER<- opcode: QUERY, status: NOERROR, id: 47470
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;google.ru.                IN      A

;; ANSWER SECTION:
google.ru.                25      IN      A      209.85.239.94

;; Query time: 25 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Wed Apr 16 21:57:16 UTC 2025
;; MSG SIZE rcvd: 54

ubuntu@ubuntu:~$ dig @1.1.1.1 skilbox.ru

<> DiG 9.20.0-2ubuntu3.1-Ubuntu <> @1.1.1.1 skilbox.ru
; (1 server found)
;; global options: +cmd
;; Got answer:
;;->HEADER<- opcode: QUERY, status: NOERROR, id: 35373
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
;; QUESTION SECTION:
;skilbox.ru.              IN      A

;; ANSWER SECTION:
skilbox.ru.              3600    IN      A      31.177.76.70
skilbox.ru.              3600    IN      A      31.177.80.70

;; Query time: 21 msec
;; SERVER: 1.1.1.1#53(1.1.1.1) (UDP)
;; WHEN: Wed Apr 16 21:57:25 UTC 2025
;; MSG SIZE rcvd: 71

ubuntu@ubuntu:~$ _
```

dig — утилита командной строки, используемая для выполнения DNS-запросов.

```
user@ubu-desk:~$ nmap -sV -v -n -p 22,23,80,443 192.168.1.11
Starting Nmap 7.80 ( https://nmap.org ) at 2025-08-13 01:45:15 UTC
```

```
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 8.9p1 Ubuntu 3 (Ubuntu Linux; protocol 2.0)
23/tcp    closed telnet
80/tcp    closed http
443/tcp   closed https
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel
```

```

ubuntu@ubuntu:~$ ethtool enp0s3
Settings for enp0s3:
    Supported ports: [ TP ]
    Supported link modes:   10baseT/Half 10baseT/Full
                           100baseT/Half 100baseT/Full
                           1000baseT/Full

    Supported pause frame use: No
    Supports auto-negotiation: Yes
    Supported FEC modes: Not reported
    Advertised link modes:  10baseT/Half 10baseT/Full
                           100baseT/Half 100baseT/Full
                           1000baseT/Full

    Advertised pause frame use: No
    Advertised auto-negotiation: Yes
    Advertised FEC modes: Not reported
    Speed: 1000Mb/s
    Duplex: Full
    Auto-negotiation: on
    Port: Twisted Pair
    PHYAD: 0
    Transceiver: internal
    MDI-X: Unknown

netlink error: Operation not permitted
Current message level: 0x00000007 (7)
drv probe link

Link detected: yes
ubuntu@ubuntu:~$ _

```

```

ubuntu@ubuntu:~$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:dd:72:29 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 83575sec preferred_lft 83575sec
    inet6 fd00::a00:27ff:fedd:7229/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 85470sec preferred_lft 13470sec
    inet6 fe80::a00:27ff:fedd:7229/64 scope link proto kernel ll
        valid_lft forever preferred_lft forever
ubuntu@ubuntu:~$ ip link show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default qlen 1000
    link/ether 08:00:27:dd:72:29 brd ff:ff:ff:ff:ff:ff
ubuntu@ubuntu:~$ ip route show
default via 10.0.2.2 dev enp0s3 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.15 metric 100
10.0.2.2 dev enp0s3 proto dhcp scope link src 10.0.2.15 metric 100
10.0.2.3 dev enp0s3 proto dhcp scope link src 10.0.2.15 metric 100
ubuntu@ubuntu:~$

```

ethtool — утилита для управления сетевыми интерфейсами Ethernet и их настройки.

ip — инструмент для управления сетевыми интерфейсами, маршрутизацией и другими аспектами сетевой конфигурации.

```

user@ubu-desk:/var/log$ journalctl -u NetworkManager

```

```

Starting Network Manager...
577]: <info> [1660158748.2092] NetworkManager (version 1.36.4) is starting... >
577]: <info> [1660158748.2213] Read config: /etc/NetworkManager/NetworkManager.conf >
577]: <info> [1660158748.2400] bus-manager: acquired D-Bus service "org.freedesktop.NetworkManager" >
577]: <info> [1660158748.2788] manager[0x55d76e631040]: monitoring kernel firmware updates >
577]: <info> [1660158748.2820] monitoring ifupdown state file '/run/network/ifupdown' >
Starting Network Manager.
577]: <info> [1660158749.2319] hostname: hostname: using hostnamed >
577]: <info> [1660158749.2319] hostname: static hostname changed from (none) to 'ubuntu' >
577]: <info> [1660158749.2331] dns-mgr[0x55d76e6112a0]: init: dns=systemd-resolved >
577]: <info> [1660158749.2343] manager[0x55d76e631040]: rfkill: Wi-Fi hardware radio set unavailable >
577]: <info> [1660158749.2343] manager[0x55d76e631040]: rfkill: WWAN hardware radio set unavailable >
577]: <info> [1660158749.2380] Loaded device plugin: NMWifiFactory (/usr/lib/x86_64-linux-gnu/libnm-device-plugin-wifi.so) >
577]: <info> [1660158749.2419] Loaded device plugin: NMTeamFactory (/usr/lib/x86_64-linux-gnu/libnm-device-plugin-team.so) >
577]: <info> [1660158749.2459] Loaded device plugin: NMAtmManager (/usr/lib/x86_64-linux-gnu/libnm-device-plugin-ata-smart.so) >
577]: <info> [1660158749.2481] Loaded device plugin: NMWwanFactory (/usr/lib/x86_64-linux-gnu/libnm-device-plugin-wwan.so) >
577]: <info> [1660158749.2509] Loaded device plugin: NMBluezManager (/usr/lib/x86_64-linux-gnu/libnm-device-plugin-bluetooth.so) >
577]: <info> [1660158749.2518] manager: rfkill: Wi-Fi enabled by radio killswitch >
577]: <info> [1660158749.2523] manager: rfkill: WWAN enabled by radio killswitch >
577]: <info> [1660158749.2527] manager: Networking is enabled by state file >
577]: <info> [1660158749.2584] settings: Loaded settings plugin: ifupdown ("/usr/lib/x86_64-linux-gnu/libnm-settings-plugin-ifupdown.so") >
577]: <info> [1660158749.2590] settings: Loaded settings plugin: keyfile (internal) >
577]: <info> [1660158749.2592] ifupdown: management mode: unmanaged >

```

NetworkManager — утилита для управления сетевыми соединениями.